

TOMAS BENCOMO

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EDUCATION

MD/PhD	University of Washington Medical Scientist Training Program Mentors: Paul Nghiem, Manu Setty, Evan Newell	2022 - present
BS	Stanford University, Computer Science Concentration in Biocomputation GPA: 3.99	2017 - 2021

HONORS AND AWARDS

Poster Prize – Fred Hutch Basic Sciences Division Retreat Third place award for best poster	2025
Oliver W. Press ARCS Endowment Award Achievement Reward For College Scientist scholarship to help fund PhD training	2024
Stanford Bio-X Summer Research Grant Funding for undergraduate research in the Dermatology department	2020
Tau Beta Pi Member Engineering honors society at Stanford	2020
Stanford VPUE Conference Grant Award to present research at Society For Investigative Dermatology Annual Meeting	2020
Mayo Clinic Summer Undergraduate Research Fellowship Fellow studying glioblastoma in the Precision NeuroTherapeutics Program	2018 & 2019
TGen Helios Scholar High school intern in the Helios Program at TGen	2016
Helios Symposium Poster Award Winner Awarded for a top 10 poster at the intern symposium	2016

RESEARCH EXPERIENCE

Graduate Student , University of Washington & Fred Hutch Cancer Center Advisors: Paul Nghiem MD/PhD, Manu Setty PhD, and Evan Newell PhD - Studied mechanisms of immunotherapy resistance using spatial multi-omics technologies - Developed novel computational approaches for spatial transcriptomics data analysis	2023 –present
Research Associate , Lee Lab, Stanford Medicine Advisor: Carolyn Lee MD/PhD - Investigated mechanisms of perineural invasion using single cell sequencing - Performed a meta-analysis of publicly available cSCC RNA-Seq datasets - Studied metabolic reprogramming caused by non-coding mutations in OXA1L	2021 – 2022
Undergraduate Researcher , Lee Lab, Stanford Medicine	2017 – 2021

Advisor: Carolyn Lee MD/PhD

- Characterized novel genes implicated in skin cancers using multi-omics data
- Examined survival data to find biomarkers for squamous cell carcinoma (SCC)
- Analyzed proteomics data to identify novel interactors for melanoma oncogene RAC1

Summer Undergraduate Research Fellow, Swanson Lab, Mayo Clinic

2018-2019

Advisor: Kristin Swanson PhD

- Analyzed MRI data to study blood brain barrier disruption in glioblastoma (GBM)
- Trained machine learning models to predict drug distribution in the brain from MRI
- Built prognostic models using radiomics data to predict patient survival

Helios Scholar, Translational Genomics Research Institute

2016

Advisor: Seungchan Kim PhD and Gil Speyer PhD

- Redesigned the lab's EDDY algorithm to use NVIDIA's CUDA API
- Helped secure \$200,000 Compute 4 the Cure grant from NVIDIA
- Achieved 500x speedup; the algorithm can now analyze previously impossible datasets

PUBLICATIONS ([GOOGLE SCHOLAR](#))

Articles (* denotes equal work)

Circulating Neoantigen- and Viral Oncoprotein-Specific CD8+ T Cells Share a Transcriptional Signature

2026

Saumya Jani*, **Tomas Bencomo***, Carolyn Shasha, Thomas Pulliam, Ana Jojic, Candice D. Church, Ted A. Gooley, David M. Koelle, Evan W. Newell, Paul Nghiem
Cancer Immunology Research

Gene expression landscape of cutaneous squamous cell carcinoma progression

2024

Tomas Bencomo, Carolyn S Lee
British Journal of Dermatology

Unravelling the landscape of skin cancer through single-cell transcriptomics

2023

Ankit Srivastava*, **Tomas Bencomo***, Ishani Das*, Carolyn S. Lee
Translational Oncology

MAB21L4 deficiency drives squamous cell carcinoma via activation of RET

2022

Ankit Srivastava, Cristina Tommasi, Dane Sessions, Angela Mah, **Tomas Bencomo**, Jasmine M Garcia, Tiffany Jiang, Michael Lee, Joseph Y Shen, Lek Wei Seow, Audrey Nguyen, Kimal Rajapakshe, Cristian Coarfa, Kenneth Y Tsai, Vanessa Lopez-Pajares, Carolyn S Lee
Cancer Research

Mutant collagen COL11A1 enhances cancerous invasion

2021

Carolyn Lee, Zurab Siprashvili, Angela Mah, **Tomas Bencomo**, Lara Elcavage, Yonglu Che, Rajani Shenoy, Sumaira Aasi, Paul Khavari
Oncogene

In Review

Extrachromosomal DNA is not common in Merkel cell carcinoma and thus unlikely to underlie its aggressive behavior

2026

Joshua Bloomstein, **Tomas Bencomo**, John C Rose, Vineet Bafna, Jens Luebeck, Paul Nghiem
Journal of Investigative Dermatology Innovations

Preprints

Epithelial tumor cells utilize mast cell-derived histamine to regulate perineural invasion 2025
Ankit Srivastava*, **Tomas Bencomo***, Chaw-Ning Lee, Angela Mah, Jasmine Garcia, Lek Wei Seow, Isoline M Donohue, Aiko J Tan, Audrey Nguyen, Tiffany Jiang, Saurabh Gombar, Lilian Phu, Pankaj Dwivedi, Christopher M Rose, RYanne Brown, Carolyn S Lee
[Biorxiv](#)

ecDNA in cutaneous squamous cell carcinoma is associated with increased nodal disease 2024
Tomas Bencomo, Amarinder Thind, Bruce Ashford, Marie Ranson, Carolyn S. Lee
[Biorxiv](#)

Conference Abstracts

Gene expression landscape of cutaneous squamous cell carcinoma progression
T. Bencomo, C. S. Lee
[Society For Investigative Dermatology 2024 Annual Meeting](#)

ecDNA in cutaneous squamous cell carcinoma is associated with increased nodal disease
T. Bencomo, A. Thind, I. Donohue, B. Ashford, M. Ranson, C. S. Lee
[Society For Investigative Dermatology 2024 Annual Meeting](#)

Mast cells orchestrate perineural invasion in aggressive epithelial cancers
A. Srivastava , **T. Bencomo** , A. Mah , J. Garcia , I. Donohue , A. Tan , L. Seow , A. Nguyen , T. Jiang , P. Dwivedi , C. M. Rose , R. Brown , S. Aasi , C. S. Lee
[Society For Investigative Dermatology 2024 Annual Meeting](#)

Metabolic reprogramming by cancer-associated mutant OXA1L promotes tumor progression and dedifferentiation
A. Srivastava, I. Donohue, A. Tan, A. Nguyen, **T. Bencomo**, J. Ye, B. Ashford, M. Ranson, D. Kashatus, C. S. Lee
[Society For Investigative Dermatology 2024 Annual Meeting](#)

Chromosomal structural variants but not extrachromosomal DNAs are common in Merkel cell carcinoma
J. D. Bloomstein , J. Luebeck , J. Rose , **T. Bencomo**, D. S. Hippe , S. Jani, V. Bafna , H. Chang , P. Nghiem
[Society For Investigative Dermatology 2024 Annual Meeting](#)

Somatic Mutation of the OXA1L 5'UTR enables Cutaneous Squamous Cell Carcinoma
Angela Mah, Jasmine Garcia, Dane Sessions, **Tomas Bencomo**, Ashley Amado, Ankit Srivastava, Carolyn Lee
[Plenary Session - Society For Investigative Dermatology 2021 Annual Meeting](#)

Molecular Profiling of Cutaneous C-Group Non-Langerhans Cell Histiocytoses
Rebekah Wieland, **Tomas Bencomo**, Carolyn Lee, RYanne Brown
[United States and Canadian Academy of Pathology 2021 Annual Meeting](#)

Sex Differences In GBM Patient Survival As A Function of Extent Of Surgical Resection and Cycles of Adjuvant Temozolomide During Standard Of Care Regimens
Julia Lorence, **Tomas Bencomo**, Haylye White, Cassandra Rickertsen, Susan Massey, Kyle Singleton, Andrea Hawkins-Daarud, Sandra Johnston, Alyx Porter, Maciej Mrugala, Bernard Bendok, Leland Hu, Joshua Rubin, Kristin Swanson
[Society for Neuro-Oncology 2020 Annual Meeting](#)

Rac1-interacting proteins are prognostic factors for melanoma survival

Marten CG Winge, **Tomas Bencomo**, M. Peter Marinkovich, Carolyn S. Lee
Society For Investigative Dermatology 2020 Annual Meeting

Sex differences in GBM treatment: an observational study

Lorence J., **Bencomo T.**, White H., Rickertsen C.R., De Leon, G. Singleton K. W., Daruud-Hawkins A., Bendok B. R., Porter A. B., Mrugala M. M., Rubin J.B., and Swanson K. R.

3rd Annual Celebration of Women's Health Research – Sexx as a Biological Variable

Sex Differences in Tumor Location Incidence in Newly Diagnosed Glioblastoma Patients

Lorence J., **Bencomo T.**, White H., Rickertsen C.R., De Leon G., Singleton K., Bendok B., Porter A. B., Rubin J. B., Swanson K. R.

Society for Neuro-Oncology 2019 Annual Meeting

Modeling the interface between non-invasive imaging and drug distribution

Ranjbar, Sara; **Bencomo, Tomas**; Jackson, Pamela R; Randall, Elizabeth; Regan, Micheal; Abdelmoula, Walid M; Lopez, Begona GC; Massey, Susan Christine; He, Lihong; Macura, Slobodan; Hu, Leland; Agar, Jefferey N; Sarkaria, Jann; Agar, Nathalie; Swanson, Kristin R.

CSBC/PS-ON 2019 Annual Investigators Meeting

The Same But Different: Identifying Distinct Imaging Ecologies in Male and Female Glioblastomas

Tomas Bencomo, Andrea Hawkins-Daarud, Kyle Singleton and Kristin R. Swanson.

CSBC/PS-ON 2018 Annual Investigators Meeting

GPU-accelerated differential dependency network analysis

Gil Speyer, Juan Rodriguez, **Tomas Bencomo** and Seungchan Kim

Euromicro International Conference on Parallel, Distributed and Network-Based Processing 2018

TALKS

Spatial interaction between tumor cells and immune hotspots defines MCC ICI resistance 4 th International Merkel Cell Carcinoma Symposium – Invited Speaker	2026
Spatial transcriptomics analysis of MCC reveals factors involved in T cell exclusion 9 th Nordic/2 nd European Merkel Cell Carcinoma Meeting	2025
Spatial transcriptomics analysis of immunotherapy response in Merkel cell carcinoma Society for Investigative Dermatology Annual Meeting	2025
Spatial transcriptomics using Xenium: An Exciting New Frontier Japanese Society of Investigative Dermatology Annual Meeting – Invited Speaker	2024
Spatial transcriptomics analysis of MCC reveals difficult to phenotype immune cell subsets 10x Seattle Single Cell and Spatial Symposium	2024
Characterizing the tumor microenvironment of MCC with spatial transcriptomics Seattle Translational Tumor Research Skin Oncology Meeting	2024
Joint modeling of cellular morphology and gene expression at single-cell resolution Fred Hutch Computational Biology Retreat: Trainee Elevator Pitch Presentation	2024

LANGUAGES

English: Native Language

Spanish: Intermediate Proficiency

TECHNICAL SKILLS

Programming: R, Python, Java, C, C++

Biostatistics: Regression modeling, machine learning, survival analysis, experimental design

Bioinformatics: Mutation profiling, bulk and single cell RNA-sequencing, spatial analysis, data visualization

Computing: Snakemake, Docker, Unix, Bash, HPC, scikit-learn, pandas, tidyverse, agentic coding

EXTRACURRICULARS

Premed Advisor Advise UW students interested in MD/PhD programs	2022 –present
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Tau Beta Pi Honors Society Board – Academic Co-Chair Organize academic development events and lead the Peer Mentorship Program	2020 – 2021
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Computer Science Peer Mentor Advise and mentor underclassmen considering a major in Computer Science	2020 – 2021
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